HISTORICAL RECREATION DOCUMENTATION

Push Engraving on Copper Plate

The Conversion on the Way to Damascus

Period-Accurate Materials, Tools, Techniques, Process, and Sources
For Presentation at Historical Reenactment Events

Prepared April 2026

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1 Object Scope and Historicity

1.1 Documentation Status

This document captures the planned and in-progress state of a hand-engraving project. The materials, tools, and methods described correspond to the engraver's intended period-accurate practice; not every step has yet been performed, and no photographs of completed components are included in this version of the documentation. The engraver has prior tool-making experience from making chasing tools and repoussé punches, and that experience underlies the forging and heat-treatment procedures described in §§2.2 and 3.1. At the time of this writing, one graver handle has been turned on the lathe from cherry (*Prunus* sp.). The graver blanks are 3/16-inch (4.8 mm) square A2 tool steel bar stock, in transit at the time of writing. The workflow will be light forging on the tang end to start a slight taper, then stock removal (grinding, filing, sanding) to finish the cross-section, point geometry, and tang profile, and finally burning the tangs into the cherry handles. Additional handles and the hammered copper plate are also still in progress and not yet documented photographically. Future revisions will incorporate photographic evidence of the gravers, handles, plate, and any pulled paper impressions, together with a dated process log. The paper should be read as period-appropriate craft documentation that will be updated as work progresses.

Version: April 2026 documentation pass; future revisions will incorporate photographic evidence and a dated process log.

1.2 Description of the Object

This project is a hand-pushed burin engraving on a hammered copper plate depicting the biblical scene of Saul's conversion on the road to Damascus (Acts 9:1–9). The engraving is executed using period-accurate tools, materials, and techniques consistent with the practice of European goldsmith-engravers of the 15th–16th centuries. The finished plate serves as both a decorative engraved object and a functional intaglio printing plate from which paper impressions can be pulled.

1.3 Skills Required

This project requires the integration of multiple distinct skills: metallurgy and tool-making (forging, hardening, and tempering steel gravers from high-carbon tool steel); woodturning (turning graver handles from dense hardwood on a lathe); metalwork (hammering and polishing the copper plate); draftsmanship (converting a painted source image into a reversed line drawing suitable for engraving); design transfer (applying the reversed drawing to the polished plate using period transfer methods); push engraving (cutting lines, hatching, cross-hatching, and stippling with a burin using hand pressure only); sharpening (maintaining precise cutting geometry on synthetic waterstones and a leather

strop); and, optionally, intaglio printing (inking, wiping, and pressing the plate to produce paper impressions).

1.4 The Biblical Subject

The Conversion of Saul on the Road to Damascus is recorded in Acts 9:1–9, with parallel accounts in Acts 22:6–11 and Acts 26:12–18 (Biblia Sacra (Vulgata Clementina) 1592). Saul of Tarsus, a persecutor of the early Christian community, is traveling to Damascus with letters authorizing the arrest of Christians. Near the city, a brilliant light from heaven strikes him to the ground and he hears a voice: "Saul, Saul, why do you persecute me?" Blinded, Saul is led into Damascus, where after three days his sight is restored and he is baptized, becoming Paul the Apostle.

The scene attracted repeated treatment by major artists across the long sixteenth century, on both sides of the Alps and in both print and paint: Lucas van Leyden's burin engraving of 1509 (Leyden 1509), Hans Baldung Grien's woodcut of c. 1515–17 (Baldung Grien 1515), Parmigianino's oil of c. 1527–28 (Parmigianino (Francesco Mazzola) 1527), Moretto da Brescia's *Conversion of Saint Paul* of c. 1540–41 on a vault of the ambulatory of Santa Maria presso San Celso in Milan (Brescia (Alessandro Bonvicino) 1540), Michelangelo's fresco in the Cappella Paolina (c. 1542–45) (Wallace 1989), and Jacopo Tintoretto's painting of c. 1544 together span the century and the principal pictorial media of the period. The National Gallery of Art, in its catalog entry on Tintoretto's *Conversion of Saint Paul*, notes that "from a religious perspective, the image of Christ winning over Saul represented divine power and specifically the victory of the Church over its enemies" and that "for artists, the story provided an opportunity to show off their skills in what often came to be treated as a battle scene" (National Gallery of Art, n.d.).

1.5 Historical Exemplars

The following are documented, extant historical exemplars demonstrating that push engraving of biblical scenes on copper was a well-established practice throughout the 15th and 16th centuries. These exemplars span multiple geographic regions, artistic traditions, and decades, providing compelling evidence that this project replicates an authentic period craft.

Exemplar 1: Martin Schongauer, Passion Series (c. 1475–1480)

A series of twelve burin engravings on copper depicting scenes from the Passion of Christ. Schongauer (c. 1450–1491) was the son of a Colmar goldsmith and the most important printmaker north of the Alps before Dürer. The canonical catalogue raisonné of his prints — Max Lehrs's 1925 volume devoted entirely to Schongauer in the *Geschichte und kritischer Katalog des deutschen, niederländischen und französischen Kupferstichs im XV. Jahrhundert* — documents the 116 surviving engravings that were distributed across Europe (Lehrs 1925). The Passion series exemplifies the practice of engraving biblical narrative scenes for devotional use. Impressions are held at the British Museum, the Victoria and Albert Museum, the Metropolitan Museum of Art, and other major print collections worldwide.

As Paula Nuttall writes for the V&A, "The Resurrection comes from a series of twelve engravings of The Passion of Christ" (Nuttall 2013).

Exemplar 2: Albrecht Dürer, The Three Master Engravings (1513–1514)

Three burin engravings on copper: *Knight, Death and the Devil* (1513), *St. Jerome in His Study* (1514), and *Melencolia I* (1514). These represent the highest technical achievement of pre-1600 push engraving. Dürer (1471–1528) learned burin technique from his goldsmith father and perfected it over decades. The engravings demonstrate the full expressive range of the burin — line, hatching, cross-hatching, and stippling — applied to complex figurative and symbolic compositions. Impressions are held in numerous major print collections worldwide, including the Metropolitan Museum of Art, the British Museum, and the Rijksmuseum. Erwin Panofsky's *The Life and Art of Albrecht Dürer* — the canonical scholarly biography — traces Dürer's command of the burin to the goldsmith training he received in his father's Nuremberg workshop, treating burin technique as a skill continuous with the goldsmith's trade rather than an independent pictorial art (Panofsky 1955). Landau and Parshall's *The Renaissance Print, 1470–1550* treats Dürer's transition from his father's goldsmith shop to independent print publication as the canonical case of the late-15th-century goldsmith-to-engraver pathway (Landau and Parshall 1994).

Exemplar 3: Maso Finiguerra, Niello Paxes (c. 1452–1464)

Engraved silver paxes with niello fill, produced for the Florence Baptistery. Finiguerra (1426–1464) was a Florentine goldsmith. Sidney Colvin, in his signed entry on Finiguerra in the 1911 *Encyclopædia Britannica*, paraphrases Cellini's praise of Finiguerra as the finest master of his day in niello engraving (Colvin 1911). His work demonstrates that engraving of religious scenes on metal — for liturgical rather than printing purposes — was standard goldsmith's practice in 15th-century Italy. Sulfur casts and paper impressions taken from his engraved plates survive, documenting the practice of recording engraved designs. Vasari, in the *Life of Marcantonio Bolognese* in the 1568 edition of the *Vite*, credits Finiguerra with taking impressions from his engraved silver plates on paper and so first revealing that engraved metal could be printed (Vasari 1998). Niello paxes are held at the Museo Nazionale del Bargello, Florence; sulfur casts at the British Museum; niello plaques at the Metropolitan Museum of Art.

Exemplar 4: Hans Baldung Grien, The Conversion of St. Paul, woodcut (c. 1515–17)

A single-sheet woodcut depicting the same biblical subject as this project. From the British Museum image, the composition shows Paul and his horse fallen, two mounted companions, and Christ in the clouds (the upper figure appears to hold a tau-shaped cross — a visual observation rather than a detail confirmed in the museum record). While a woodcut rather than an engraving, this demonstrates that the Conversion of Paul was an established subject for printmakers in the early 16th century. The bold composition and dramatic treatment of the scene provide a period model for the subject. The British Museum collection record describes it as "the Conversion of St Paul; St Paul and his horse on the ground, struck and

blinded by the bright light which is set against a dark sky and sent by Christ" (Baldung Grien 1515).

Exemplar 5: Lucas van Leyden, The Conversion of St. Paul, engraving

A burin engraving on copper depicting the same subject. Lucas van Leyden (1494–1533) was one of the leading engravers of the Low Countries. This exemplar is directly relevant because it is both an intaglio burin engraving and depicts the identical biblical scene, demonstrating that the Conversion of Paul was an established subject for copper-plate engravers (Leyden 1509).

Additional Evidence of Use and Purpose

Engraved devotional prints served documented purposes in the 15th–16th centuries: as affordable alternatives to painted devotional images for middle-class buyers; as personal devotional aids that could be inserted into prayer books; as commercial products sold individually or in series; and as pattern sources for other artists working in painting, sculpture, and other media. The V&A notes that "printed images on paper were to some extent inexpensive alternatives to paintings" (Nuttall 2013). Landau and Parshall observe that serving as a pattern for other artists working in painting, sculpture, and the decorative arts was itself one of the primary purposes for which many Renaissance prints were produced (Landau and Parshall 1994). Goldsmiths also engraved directly onto liturgical objects (chalices, paxes, reliquaries) for ecclesiastical use, and onto secular objects (armor, jewelry, belt fittings) for decorative purposes.

2 Materials

Every material used in this project is discussed below with its historical basis, specific period evidence, and — where modern substitutions are made — a comparative analysis explaining the deviation and its justification.

2.1 The Plate: Copper

Historical Material

Copper was the universal plate material for intaglio engraving from the 1430s through the end of the period and beyond. It was chosen because it is soft enough to cut cleanly with hand tools, hard enough to hold fine detail, does not rust, and can produce many impressions before wearing. Period copper plate was produced by hammering ingots or sheet to the desired thickness and then polishing to a mirror finish. Landau and Parshall's chapter on the beginnings of engraving documents copper as the standard substrate of the fifteenth-century intaglio shops, chosen for its combination of workability under hand tools and resistance to corrosion (Landau and Parshall 1994), and Hind gives the same account in the opening chapters of his *History of Engraving and Etching* (Hind 1963).

Period vs. Modern Copper

Period copper plate was hammered, not rolled. Modern commercially available copper sheet is cold-rolled, producing a uniform grain structure that differs from the work-hardened pattern of period hammered plate. Landau and Parshall's study of late-15th- and early-16th-century intaglio practice documents the production and preparation of period copper plates by the goldsmith-engravers of the Upper Rhine in detail, noting that hammered plate was the universal substrate of the early intaglio shops (Landau and Parshall 1994). Hind's *History of Engraving and Etching* gives the same account in chs. I–II (Hind 1963).

Decision

For this project, the copper plate is hand-hammered from copper sheet and polished to a mirror finish, replicating the period preparation method. If modern cold-rolled copper sheet is used as a starting point, this substitution is justified by availability and cost — the functional difference is in grain structure only, and the rolled sheet can be further worked with a planishing hammer to approximate hammered plate. The material itself (copper) is identical to the period material.

2.2 Tool Steel: Gravers

Historical Material

Period gravers were made from high-carbon steel, hardened and tempered by the engraver or a toolsmith. Theophilus Presbyter's *De Diversis Artibus* (c. 1100–1120), the oldest surviving complete treatise on metalworking crafts, describes the making and heat treatment of the full array of the goldsmith's steel bench tools in Book III, including a dedicated chapter (ch. XIV, "De Ferris Incisoriis" / "Of Cutting Instruments") on the hand-held cutting irons with which "gold is cut, and silver, or thick copper":

> Cutting instruments are also made broad and even, of such a size that they may be held in the whole hand, and that they may rise above the hand: they even extend below the hand broad, thin and sharp. Many of these are made large and small; with these gold is cut, and silver, or thick copper.

(Theophilus, *De Diversis Artibus*, Book III, ch. XIV, "Of Cutting Instruments," Hendrie 1847 translation (Theophilus Presbyter 1847).) A few chapters later, on the making of files — close relatives of the graver, also forged from high-carbon steel, hardened by quenching, and finished with cut teeth — Theophilus is explicit that the stock material must be the highest grade of steel available to a medieval workshop:

> Files are made from pure steel, large and moderately sized, triangular and round. Others also are made which are stronger in the middle, from soft iron inside, but outwardly they are covered with steel.

(Theophilus, *De Diversis Artibus*, Book III, ch. XVII, "Of Files," Hendrie 1847 translation (Theophilus Presbyter 1847).) The same "pure steel" standard documented here for files applies across the small hand tools of the goldsmith's bench that Book III describes — including the cutting irons of ch. XIV and the grooving irons of ch. XX, all forged from the same high-grade stock. Commercially available gravers in the period would have been forged from bloomery or crucible steel with a carbon content sufficient to harden by quenching — roughly equivalent to modern W-1 (water hardening) or O-1 (oil hardening) tool steel, which are plain high-carbon steels (Theophilus Presbyter 1979).

Comparative Analysis: A2 Tool Steel vs. Period Steel

This project uses 3/16-inch (4.8 mm) square A2 tool steel bar stock — an air-hardening tool steel — shaped by a hybrid workflow: light forging on the tang end to start a slight taper, followed by stock removal (grinding, filing, sanding) to finish the cross-section, point geometry, and tang profile. This avoids the more demanding work of forging the full cross-section down from a larger bar while still using the period operation of hot-tapering the tang. A2 differs from period carbon steel in that it contains chromium and molybdenum additions that allow it to harden in air rather than requiring a liquid quench. The functional result — a hardened steel cutting tool at approximately Rockwell C 58–62 — is identical. The advantages of A2 for this application are: reduced risk of quench cracking on the thin cross-sections of gravers; minimal distortion during heat treatment; and the engraver's

existing familiarity with the material from prior toolmaking experience (chasing chisels and repoussé punches).

The substitution is justified on practical grounds: the cutting performance, hardness, and edge-holding characteristics are functionally equivalent to period carbon steel. The tool's behavior under the burin is functionally equivalent to a period graver for cutting copper at this hardness range; the differences in chip behavior between A2′s alloyed steel and a plain-carbon period blade are subtle and would only matter for the most demanding fine-line work. For maximum period authenticity, O-1 or W-1 tool steel could be used instead, as these are plain carbon steels closer in composition to period bloomery steel, requiring oil or water quench respectively. Old file steel is another documented traditional source material that closely approximates period carbon steel. Experienced engravers report making functional gravers from O-1, W-1, drill rod, and old file steel, all of which are plain high-carbon steels (Roki1967 (forum user) 2007; DocBernard (forum user) 2023).

2.3 Graver Handles: Dense Hardwood

Historical Material

The traditional material for graver handles is a dense, fine-grained hardwood that does not split under the tang and does not deform under the heel of the palm. Modern reference works on the burin name boxwood (*Buxus sempervirens*) as the species most often used in surviving and contemporary handles, though the period treatises that describe the burin in detail rarely specify the wood. Bosse, in his 1645 *Traicté*, refers only to a "bois ferme & non sujet à s'esclatter" — a firm wood not prone to splitting — when he describes fitting handles to small engraving tools (Bosse 1645), and Theophilus, in Book III of *De Diversis Artibus*, describes the cutting irons of the goldsmith's bench as "fitted in their handles" without naming the species (Theophilus Presbyter 1847). The shape, by contrast, is well-attested across both period and modern descriptions: the handle is mushroom-shaped with one side cut flat, approximately 1.5–2 inches (4–5 cm) in length, providing an ergonomic palm grip that positions the shaft low to the work surface. The 1911 *Encyclopædia Britannica*'s entry on Line-Engraving describes the burin as fitted with "a handle rather like a mushroom with one side cut away" (*Line-Engraving* 1911).

Decision

Handles for this project are turned on a lathe from cherry (*Prunus* sp.). Cherry is dense, fine-grained, and resistant to splitting under the tang — meeting Bosse's *"bois ferme & non sujet à s'esclatter"* criterion exactly. Cherry and other European fruitwoods (pear, apple, plum) were standard period turnery materials in fifteenth- and sixteenth-century European workshops, used widely for small tool handles, instrument parts, and printing blocks; they are functionally equivalent to boxwood for this application, with the same density range and resistance to splitting. At the time of this writing, the first cherry-wood handle has been completed; each graver will have its own dedicated handle with the tang burned into a drilled starter hole — no ferrule, no set screw, no glue. The burning-in method (heating the tang to a low red and pressing it into the wood, so the steel scorches the wood to its own

exact profile) is the traditional pre-modern fit for small steel hand tools and is described in §3.1.4. This replicates the documented period practice in which engravers maintained a set of individually handled tools.

2.4 Sharpening Stones

Historical Material

Period European engravers sharpened on locally available natural hones — fine-grained sandstones, slates, and schistose oilstones quarried in the Rhineland, the Vosges, the Italian Alps, and other European sources. A leather strop dressed with abrasive powder (rouge or tripoli) was used for final polishing. The general workflow — coarse stone, then finer stone, then leather strop — has been continuous from the period to the present, even as the specific stone sources have changed.

Decision and Substitution

This project uses synthetic Japanese waterstones in a 1000 / 4000 / 8000 grit progression, finished with a leather strop dressed with chromium-oxide polishing compound. **Synthetic waterstones are a substitution.** Modern synthetic waterstones — bonded aluminum-oxide or silicon-carbide abrasives in a friable resin matrix — were not available to a 15th- or 16th-century European engraver, who would have used local natural hones of the kinds described above. Waterstones cut faster than natural hones, expose fresh abrasive as they wear, and are available at extremely fine grits (8000 and above) that produce a polish equivalent to the finest natural novaculite or period European hone. For burin work specifically, the leather strop with chromium-oxide compound is what brings the cutting edge to its final mirror polish, and that final step is the same regardless of which stones precede it. The functional result — a mirror-polished face and heel on a burin sharp enough to lift a clean copper curl — is the same edge a period engraver would have achieved with their own local stones and strop. Ganoksin describes the modern workflow, which mirrors the period sequence: "Grinding is usually done on an oilstone, working in a circular stroke, first on a coarse stone and then on a finer one. […] Follow this with a few strokes on a hard leather with a polishing powder to create a mirror bright finish" (Ganoksin, n.d.). No diamond plates, ceramic stones, or powered sharpening systems are used.

2.5 Design Transfer Medium

Historical Method

Period engravers transferred designs by tracing from a paper drawing using a stylus. The transfer medium was a dark or colored powder applied to the back of the drawing, which left faint guide lines on the plate when traced over with a pointed tool. Abraham Bosse, in his 1645 *Traicté*, devotes a chapter ("Manière de s'apprester pour desseigner, contre tirer ou calquer son dessein sur la planche") to exactly this operation, describing two equivalent arrangements: the drawing is either rubbed directly on its reverse with powder, or laid over

a separate sheet of paper prepared with powder, then fixed to the plate with soft wax and traced over with a "pointe à calquer" so that the contours of the design are transferred onto the plate below (Bosse 1645). Bosse writes of the drawing "on bon papier" whose verso is "rougy [...] en respandant dessus et la frottant avec un petit linge, en sorte qu'il soit bien également rouge par tout" — uniformly reddened with powdered sanguine — and then of passing the calquing point "sur tous les contours des figures qui le composent, en l'appuyant assez fort et également [...] et cela fait vous devez sçavoir que tous les contours de vostre dessein [...] seront marquez empraints ou calquez [...] de la planche." Bosse's method uses sanguine (powdered red chalk), but the technique is the same back-of-the-drawing transfer that survives in workshop practice, substituting lamp-black soot for the red powder when a dark line on a bright polished surface is preferred.

Decision

Lamp black (soot from a candle or oil lamp) rubbed onto the back of the reversed drawing, then traced with a steel stylus. Modern carbon paper is a functional equivalent but not strictly period; lamp black transfer is the documented period method and is used for this project.

2.6 Ink and Paper (If Printing)

If paper impressions are pulled from the finished plate: period printing ink was oil-based, made from lampblack ground in linseed oil. Period paper was handmade laid paper (showing chain lines and laid lines from the papermaking mould), dampened before printing. Wove paper was not invented until the 18th century and would be anachronistic. Modern handmade laid paper is available from specialty suppliers and is functionally identical to period paper.

3 Methods and Tools

Every method and tool used in this project is described below with its historical basis. Where any deviation from strictly period practice occurs, it is identified, described, justified, and compared to the period method.

3.1 Tool Making

Shaping the Gravers

Each graver is shaped from a length of 3/16-inch (approximately 4.8 mm) square A2 tool steel bar stock, cut to roughly 100–115 mm (4–4.5 inches). The workflow is a hybrid of light forging and stock removal: the tang end is first brought to a slight taper by hot working — heated to a forging temperature and tapped down on the anvil over a few passes to start the profile — and the rest of the shaping is then done cold, by grinding wheel, files, and progressively finer abrasive papers. The cutting end is brought to its square or lozenge cross-section and point geometry by stock removal, and the forged tang taper is dressed and finished by file and abrasive paper. Both halves of this workflow are documented period operations: forging to start a tang taper is the universal approach to fitting tools into burned-in handles, and Theophilus discusses files at length in Book III of *De Diversis Artibus*, including the chapter "Of Files" already quoted in §2.2 (Theophilus Presbyter 1847). The filed-and-fitted sequence Theophilus describes in chapter XX (quoted in §3.1.2 below) names exactly the operations used here: file the iron to shape, fit it into the handle, then heat-treat. A modern French engraver documents an analogous workflow — old file steel cut to length, heat-treated at 850°C, quenched in water, and fitted to a lathe-turned wooden handle — demonstrating that the same sequence remains the standard practitioner approach today (Roki1967 (forum user) 2007).

Heat Treatment

The shaped blanks are hardened by heating to non-magnetic temperature (approximately 1475°F / 800°C for plain carbon steels; higher for A2 due to its alloy content) and air-cooling (A2) or quenching in oil or water (O-1 / W-1). For A2 specifically, the austenitizing temperature is approximately 1750–1800°F / 955–980°C; the steel is then cooled in still air. They are then tempered to a straw color to achieve approximately Rockwell C 58–62. This hardness range allows the tool to hold a sharp edge while cutting soft metals (copper, brass, silver) without being so brittle that the tip chips readily. Period heat treatment used the same principles — heating to critical temperature and quenching — with color judgment as the primary means of gauging temperature, exactly as practiced here. Theophilus, in Book III of *De Diversis Artibus*, gives the simplest possible recipe for hardening the small cutting tools of the goldsmith's bench:

> Grooving irons are likewise tempered in this manner. When they have been filed and fitted in their handles, their end is placed in the fire, and presently, when it has begun to glow, it is taken out and quenched in water.

(Theophilus, *De Diversis Artibus*, Book III, ch. XX, "Of Tempering Iron," Hendrie translation (Theophilus Presbyter 1847).) The procedure described — heat the working end to a glow, quench, then return to the fire briefly to draw the temper — is the same operation performed in this project, with the modern conveniences of a thermocouple and a tempering oven where Theophilus and his contemporaries relied on color and experience (Theophilus Presbyter 1979). Ganoksin's guidance reflects the same logic: "gravers are sold in their hardened state and any grinding should be done in such a way that this temper is preserved. Touch the tool lightly to the grinding wheel, dipping it frequently into water to prevent overheating. If the steel starts to show color the temper has been compromised" (Ganoksin, n.d.).

Sharpening Geometry

The cutting face is ground at 45° on the diagonal, with 15° heels (relief bevels on the underside). For hand pushing specifically, a lower face angle of 35–40° with a longer heel is preferred, as this geometry requires less force and provides more control than the steeper angles used with modern pneumatic tools. Sharpening is done on synthetic Japanese waterstones in a 1000 / 4000 / 8000 grit progression, followed by a leather strop dressed with chromium-oxide compound (see §2.4). The face and heel surfaces must be flat and polished to a mirror finish. The Antique Firearms Restoration Blog describes the standard starting geometry: "the main face is cut at 45 degrees on the diagonal and the heels are cut at 15 degrees. These angles can be varied according to personal preferences, but 45/15 are a good starting point" (Cathro, n.d.). Experienced hand-push engravers recommend lower face angles of 35–40° and longer heels for hand work (sam (forum user) 2015).

Handle Turning and Tang Fitting

Handles are turned on a lathe from cherry (*Prunus* sp.; see §2.3 for the period justification) to the documented period mushroom-with-flat-bottom shape. A small starter hole is drilled at a slight angle from the flat face. The tang is then **burned into the handle**: the tapered end of the tang is heated to a low red and pressed firmly into the starter hole. The hot steel scorches the wood to the tang's exact profile, leaving carbonized walls that grip the steel without slop and seat the shaft at the precise angle the engraver wants. No ferrule, no set screw, no glue — the burned fit alone holds the tool together for a working lifetime. Burning-in is the traditional pre-modern method for fitting wooden handles to small steel hand tools (chisels, gouges, files, awls, gravers); it produces a tighter and more durable fit than a cold press into a precisely-drilled hole, and it is the method Bosse's "bois ferme" criterion implicitly assumes — a splitty wood would crack rather than scorch.

3.2 Plate Preparation

The copper sheet is cut to size, hammered to working thickness if needed (period plates were hammered, not rolled), and polished to a scratch-free mirror finish. Period polishing was done progressively with finer abrasives and burnishing. For this project, the plate is polished using progressively finer grades of abrasive, finishing with a polishing compound — functionally identical to period practice, though modern abrasive papers may be used in place of period abrasive stones as a convenience substitution with no functional difference.

3.3 Design Preparation and Transfer

The source image is Moretto da Brescia's *The Conversion of Saint Paul* (c. 1540–41), painted on a vault of the ambulatory of Santa Maria presso San Celso in Milan, one of the sixteenth-century treatments of the subject enumerated in the iconographic tradition discussed in §1.4 (Brescia (Alessandro Bonvicino) 1540). The painting is studied and interpreted as a line drawing. This is a creative act of translation — the engraver must decide how to represent painted tonal values as line work (hatching, cross-hatching, stippling). This process of converting paintings into engraved line compositions was standard period practice; Marcantonio Raimondi (c. 1480–1534) built his career on engraving reproductions of Raphael's paintings. Lisa Pon's monograph *Raphael, Dürer, and Marcantonio Raimondi: Copying and the Italian Renaissance Print* documents the systematic translation of Raphael's compositions into engraved reproductions in Marcantonio's workshop, and treats this reproductive practice as central to the formation of the sixteenth-century Italian print trade (Pon 2004).

The line drawing is reversed (since the print will be a mirror image of the plate) and transferred to the polished plate using lamp black on the back of the drawing, traced with a steel stylus. Dürer is documented as having worked from reversed preparatory drawings when planning his engravings; Panofsky's biography discusses Dürer's preparatory practice in detail (Panofsky 1955).

3.4 Engraving Technique

Grip and Cutting

The burin is held at approximately 30° to the plate surface. The mushroom handle is cradled in the palm and the index and middle fingers guide the shaft. Pressure is applied both downward and forward. The tool cuts into the metal, forming a chip and pushing a thin curl of metal ahead of it. This is push engraving — no hammer, no pneumatic assist, hand pressure only. Antony Griffiths, in the standard British Museum handbook on prints and printmaking, describes the same modern push-engraving grip and the basic burin-and-chip mechanics that have remained unchanged from the fifteenth century to the present (Griffiths 1996).

Plate Rotation

The plate rests on a leather sandbag cushion and is rotated with the non-cutting hand to produce curves. The cushion is the same simple device that Abraham Bosse describes for his own workshop in 1645, in the chapter on holding and managing the burin:

> And to this end you must make yourself a small cushion of fairly stout leather, somewhat in the shape of the pincushions that women and girls use to hold needles and pins, half a foot square in width and three or four inches in height, filled with rather fine sand; and you shall set this cushion upon a table firmly fixed in place. Then you shall lay your plate upon this cushion, so that you may turn it according as the lines and hatchings shall require.

(Translated from Bosse, *Traicté des manieres de graver en taille dovce svr l'airin*, Paris, 1645, in the chapter "Briève maniere pour sçavoir tenir, manier & aiguiser un Burin." Cited from the Duke University Libraries copy digitized by the Internet Archive (Bosse 1645).) A. Hyatt Mayor, in *Prints and People*, observes that Schongauer was the first engraver to curve parallel lines by rotating the plate against a steady burin (Mayor 1971, p. 130) — a technique that, as Bosse confirms two centuries later, depends entirely on the sand-filled leather cushion holding the plate steady but free to swing. The Diderot Encyclopédie plates clearly illustrate the same arrangement (Diderot and d'Alembert 1767).

Line Vocabulary

All tonal variation is achieved through line character and density. The line vocabulary consists of: pure line (continuous clean strokes for contours); hatching (parallel lines of varying spacing and depth for shading); cross-hatching (crossed sets of parallel lines for deeper shadows, pioneered by Schongauer and perfected by Dürer); and stippling (dots and short flicks for texture, documented from the late 15th century). Wendy Thompson describes Schongauer's technique: he "used a great variety of strokes — from the long, sinuous lines that create the beard of the saint and the curling fur of one of the demons, to the short flecks of the saint's coarsely woven robe; he also made use of crosshatching in the deepest shadows to model the forms" (Thompson 2003).

Workflow Sequence

The period engraving workflow, documented across multiple sources:

1. Design preparation;
2. Plate preparation and polishing;
3. Design transfer in reverse;
4. Contour engraving (main outlines first);
5. Shading (hatching, cross-hatching);
6. Textures and details (stippling, fine work);
7. Cleaning and finishing (removing burrs and metal curls);
8. Printing if desired (inking, wiping, pressing) (Curtidos Cabezas, n.d.).

3.5 Deviations from Period Methods

The following deviations from strictly period practice are made in this project, with justification:

A2 tool steel instead of period carbon steel Justified by the engraver's existing familiarity with this material from prior toolmaking. The functional result (hardened steel graver at Rc 58–62) is identical. See §2.2 for full comparative analysis.

Synthetic Japanese waterstones for sharpening Used in place of period European natural hones. The functional result (a mirror-polished face and heel on the burin, finished on a leather strop) is the same edge a period engraver would have produced. See §2.4 for full discussion.

Modern abrasive papers for plate polishing Used as a convenience alongside traditional polishing methods. The functional result (mirror-polished copper surface) is identical to period preparation. Period engravers used progressively finer natural abrasive stones and burnishing to achieve the same result.

Electric lighting Period engravers worked by natural light or candlelight. Modern lighting is used for practical reasons at indoor events. This has no effect on the object itself.

No pneumatic or power-assisted tools are used. No diamond or ceramic sharpening equipment is used. No modern quick-change handle systems are used. Aside from the substitutions enumerated above, the core operation — pushing a hand-held steel burin into a polished copper plate to lift a curl of metal — is unchanged from the period practice documented by Theophilus, Bosse, and the surviving plates of Schongauer and Dürer.

4 Research and Supporting Evidence

4.1 Types of Evidence

This project's documentation draws on multiple types of evidence:

Primary written sources Period treatises by practicing craftsmen describing tools, materials, and techniques from firsthand experience. These include Theophilus Presbyter's *De Diversis Artibus* (c. 1100–1120) (Theophilus Presbyter 1979), Benvenuto Cellini's *Trattati dell'oreficeria* (1560s) (Cellini 1967), and Abraham Bosse's *Traité des manières de graver en taille douce* (1645, documenting practices unchanged from the 16th century) (Bosse 1645).

Extant objects Surviving engraved copper plates, prints pulled from those plates, engraved niello silverwork, and engraved decorative metalwork held in major museum collections (Metropolitan Museum of Art, British Museum, Rijksmuseum, Victoria and Albert Museum, Bargello). These provide direct physical evidence of period materials, techniques, and subjects.

Visual documentation Abraham Bosse's engraved illustrations of the engraver's workshop (1643–45) (Bosse 1645), the Diderot Encyclopédie plates on "Gravure en taille-douce" (c. 1767–1771, documenting practices unchanged from the 16th century) (Diderot and d'Alembert 1767), and period depictions of goldsmith workshops.

Modern scholarly analysis Museum publications, art-historical scholarship, and technical analyses of period engravings. Key sources include the Metropolitan Museum's Heilbrunn Timeline of Art History essays on engraving (Thompson 2003), the National Gallery of Art's exhibition catalogs (National Gallery of Art, n.d.), and the standard art-historical treatments of the Renaissance print by Landau and Parshall (Landau and Parshall 1994), Hind (Hind 1963), and Griffiths (Griffiths 1996).

Practitioner knowledge Modern hand engravers who practice period techniques, documented in specialized forums (the Engraver's Cafe (Roki1967 (forum user) 2007; DocBernard (forum user) 2023; sam (forum user) 2015; kayscheible (forum user) 2017)) and instructional materials. While modern, these practitioners preserve and transmit technical knowledge about hand-tool geometry, sharpening, and cutting technique that is directly continuous with period practice.

4.2 Discussion of Sources and Historic Context

The practice of push engraving on copper for the production of devotional prints emerged from the European goldsmithing trade in the 1430s. Landau and Parshall locate the origin of intaglio engraving along the Upper Rhine in southwestern Germany in the 1430s, growing out of goldsmiths' decorative techniques on metal, and trace the goldsmith-to-engraver transition and the material practices of the early intaglio shops in detail (ch. 1) (Landau and Parshall 1994); the same trajectory is described in Hind's classic *A History of Engraving*

and Etching (chs. I–II) (Hind 1963). Landau and Parshall further note that the notable engravers of the fifteenth century were either goldsmiths themselves (Master E.S.) or the sons of goldsmiths (Martin Schongauer, Albrecht Dürer), and that the technique had long been used by goldsmiths on their own metal pieces before any impressions were taken from them (Landau and Parshall 1994). Antony Griffiths's *Prints and Printmaking* likewise locates the origins of European engraving in the workshop practices of fifteenth-century goldsmiths (Griffiths 1996).

The Italian tradition developed simultaneously from niello work. The standard scholarly catalogue of early Italian engraving — Levenson, Oberhuber, and Sheehan's *Early Italian Engravings from the National Gallery of Art* — documents how the very earliest Italian engraved prints were impressions taken from silver plates intended for niello, before the niello fill was added (pp. 528–529) (Levenson et al. 1973); Landau and Parshall give the same account (pp. 26, 67–68) (Landau and Parshall 1994). Vasari, in the *Life of Marcantonio Bolognese* in the 1568 edition of the *Vite*, credits Maso Finiguerra (1426–1464) with the invention of copperplate printmaking from impressions of niello plates (Vasari 1998); A. Hyatt Mayor's *Prints and People* gives the modern reassessment of this account, showing that Finiguerra was one of several Florentine goldsmiths working in the same niello-print tradition rather than its sole inventor (Mayor 1971).

The earliest and most important primary source on medieval metalworking tools and techniques is Theophilus Presbyter's *De Diversis Artibus*. C. R. Dodwell's critical edition introduces the *De Diversis Artibus* as the most exhaustive surviving account of medieval craft technique, written by a Benedictine monk who was himself a practicing metalworker (Theophilus Presbyter 1986). The identification of Theophilus with Roger of Helmarshausen — a goldsmith whose surviving metalwork demonstrates the techniques described in the text — was first proposed by Albert Ilg in 1874 and developed by Dodwell in his 1961 critical edition; Dodwell himself treats the identification with caution, writing that the Benedictine monk-author "may have been" Roger (Theophilus Presbyter 1986).

Abraham Bosse's 1645 *Traité des manières de graver en taille douce* is the first published treatise specifically on engraving (Bosse 1645). While published just after the period boundary, it documents practices and tools that were functionally unchanged from the previous century, as the burin, the sandbag cushion, the sharpening methods, and the printing process it illustrates are all attested in earlier sources and extant objects. The treatise was widely translated and remained the standard reference for over a century.

5 Execution, Complexity, and Creativity

5.1 Project Goals

The following goals were established for this project:

1. Forge and heat-treat a set of push engraving gravers from tool steel, replicating the period practice of engravers making their own tools.
2. Turn period-accurate mushroom handles from dense hardwood.
3. Prepare a copper plate using period-appropriate methods.
4. Convert Moretto da Brescia's c. 1540–41 *Conversion of Saint Paul* (Santa Maria presso San Celso, Milan) (Brescia (Alessandro Bonvicino) 1540) into a reversed line drawing suitable for engraving, exercising personal artistic interpretation in the translation from painted tonal values to line work.
5. Transfer the design and engrave the plate using push engraving only, employing the full period line vocabulary (line, hatching, cross-hatching, stippling).
6. Produce a finished engraved plate that functions both as a decorative object and as a printing plate capable of producing paper impressions.
7. Document every material, tool, and technique decision with reference to period sources.

5.2 Personal Interpretation

This project is not a direct copy of an existing engraving. The source material is a painting, and the conversion of a painted image into engraved line work requires substantial personal artistic interpretation. The engraver must decide: which painted values become lines and which become negative space; how to represent color and tonal gradation using only black-and-white line vocabulary; how to simplify or adapt the composition for the constraints of the burin; and how to manage the reversal of the image. This process of converting paintings to engravings was itself a recognized and respected skill in the period — Marcantonio Raimondi was celebrated specifically for his ability to translate Raphael's paintings into copper plate engravings, a workshop practice documented in detail in Lisa Pon's *Raphael, Dürer, and Marcantonio Raimondi* (Pon 2004).

5.3 Challenges and Solutions

Challenge 1: Learning Push Engraving as a New Skill

Push engraving requires a fundamentally different hand motion from the engraver's prior experience with chasing and repoussé (hammer-driven) and woodblock carving (knife-driven). The burin is pushed with palm pressure, requiring development of new muscle memory for control of line depth, width, and curvature. *Solution:* extensive practice on scrap copper with straight lines, curves, and basic scroll elements before beginning the finished

plate, following the progression recommended by experienced engravers (kayscheible (forum user) 2017).

Challenge 2: Converting Painted Tonal Values to Line Work

A painting has continuous tonal gradation; an engraving has only lines. Every area of the composition must be interpreted as some combination of line density, and there is no way to correct major compositional decisions once cutting has begun. *Solution:* extensive preparatory drawing in pencil, working through the tonal interpretation on paper before committing to the plate. Multiple studies at different simplification levels to determine the appropriate level of detail for the engraver's current skill.

Challenge 3: Image Reversal

The engraved image must be cut in reverse so that it prints correctly. This requires the engraver to work with a mirrored version of the design, which can be disorienting. *Solution:* prepare the reversed drawing carefully before transfer, and work from the transferred lines on the plate rather than from memory or a non-reversed reference.

Challenge 4: Sharpening to Precise Geometry

A burin that is not perfectly sharpened with correct face and heel geometry will dig in, skip, or produce ragged lines. Maintaining consistent geometry during resharpening (which is required frequently) is critical. *Solution:* develop a consistent sharpening routine using the natural stone and strop, with the thumbnail test to verify sharpness before each session. Practice resharpening until it becomes reliable and fast.

Challenge 5: Tool Making with Precision Geometry

While the engraver has prior experience making chasing tools, gravers require more precise geometry at the cutting face. The face angle, heel angle, and point geometry must all be correct for the tool to function properly. *Solution:* forge multiple blanks and practice grinding geometry on the waterstones before committing to the final tool set.

5.4 Future Avenues of Exploration

This project opens several avenues for further development:

Niello fill Adding niello (a sulfur-silver-copper-lead compound) to the engraved lines to create a high-contrast decorative object in the Florentine goldsmith tradition. This would add a new skill (niello preparation and application) and connect the project to the Italian engraving tradition of Finiguerra and his contemporaries. The standard scholarly catalogue of early Italian engraving — Levenson, Oberhuber, and Sheehan's *Early Italian Engravings from the National Gallery of Art* — treats the Florentine niello-print tradition in detail (Levenson et al. 1973), and Cellini's *Trattati* covers sixteenth-century goldsmiths' niello practice firsthand from a working Florentine shop (Cellini 1967).

Intaglio printing Building or obtaining a simple screw press and pulling editions of paper prints from the plate. Period presses were simple wooden constructions, and a period-accurate press is itself a viable reenactment project.

Engraving directly on liturgical or decorative objects Moving from flat plates to three-dimensional objects such as chalice bands, paxes, or belt fittings, replicating the goldsmith's decorative practice that predated printmaking.

Sulfur casting Making sulfur casts from the engraved plate to record the design, replicating the documented practice of Finiguerra and other Renaissance goldsmiths as recorded by Vasari (Vasari 1998) and reassessed in A. Hyatt Mayor's *Prints and People* (Mayor 1971).

More complex compositions Progressing to multi-figure scenes with full cross-hatched shading in the manner of Dürer's master engravings.

Period pigment preparation Making printing ink from lampblack ground in linseed oil, and preparing laid paper, to produce fully period-accurate prints.

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